

Unitary representations

1

$$G = \mathbb{R}^N$$

$$\Gamma = \mathbb{Z}^N$$

$$\Gamma \backslash G = \mathbb{T}^N := \mathbb{R}^N / \mathbb{Z}^N$$

$$G \curvearrowright \mathcal{H} = L^2(\mathbb{T}^N, d\theta)$$

$$(t, \varphi) \mapsto (\pi(t)\varphi: x \mapsto \varphi(x+t))$$

$$SL_N(\mathbb{R}) = \{ N \times N \text{ matrix} \\ \text{with } \det = 1 \}$$

$$\Gamma$$

$$\Gamma \backslash G$$

$$L^2(\Gamma \backslash G; d\mu)$$

$$\underline{SL_N(\mathbb{R})}$$

Lie gr. locally cpt top group

$$SL_N(\mathbb{R}) = \{ g \in M(N, \mathbb{R}) \mid \det g = 1 \}$$

$$g \text{ left action } \lambda_g(x) = gx$$

$$\text{right action } \ell_g(x) = xg^{-1}$$

(Haar measure) essentially unique

lem: 1) $\exists$ borel measure ν on G , finite on compact, s.t.

$$(\lambda_g)_* \nu = \nu$$

$$2). \forall g \in G, (\ell_g)_* \nu = \nu$$

pf. let λ be Lebesgue measure on $M(N, \mathbb{R})$

λ is λ_g, ℓ_g -inv. for $g \in G$, change variable

1). Define $\nu(A) = \lambda \left\{ x \in M(N, \mathbb{R}) \mid \frac{x}{(\det x)^{1/N}} \in A \right\}$ (2)
 $1 \leq \det x \leq 2$

$$\nu(gA) = \lambda \left\{ x \in M(N, \mathbb{R}) \mid \frac{x}{(\det x)^{1/N}} \in gA \right\}$$

$$= \lambda \left\{ x \mid \det x \leq 2, \frac{g^T x}{\det(g^T x)^{1/N}} \in A \right\}$$

$$= \lambda \left\{ y \mid \det y \leq 2, \frac{y}{\det(y)^{1/N}} \in A \right\} = \nu(A)$$

right invariant is similar

□

Metric: (ad hoc) if we use Lie theory, more simple?

$x \in M(N, \mathbb{R})$. operator norm $\|x\| = \sup_{v \neq 0} \frac{\|xv\|}{\|v\|}$, $\|\cdot\|$ euclidean

$$\forall x, y \in G. \quad d_G(x, y) = \log(1 + \|1 - x^T y^\circ\|) + \log(1 + \|1 - y^T x^\circ\|)$$

Triangle inequality

$$\log(1 + \|1 - xz^{-1}\|)$$

$$\leq \log(1 + \|1 - xz^{-1} + 1 - zy^{-1} - (1 - xz^{-1} - zy^{-1} + xy^{-1})\|)$$

$$\leq \log(1 + \|1 - xz^{-1}\| + \|1 - zy^{-1}\| + \|1 - xz^{-1} - zy^{-1} + xy^{-1}\|)$$

where $1 - xz^{-1} - zy^{-1} + xy^{-1} = (1 - xz^{-1})(1 - zy^{-1})$

$$= \log(1 + \|1 - xz^{-1}\|) + \log(1 + \|1 - zy^{-1}\|)$$

$$\Rightarrow d_G(x, y) \leq d_G(x, z) + d_G(z, y)$$

Then $d_G(gx, gy) = d_G(x, y)$ left G -invariant metric

Let Γ be a discrete subgroup of G . Then $\exists \varepsilon > 0$. (3)

s.t. $B_G(e, 2\varepsilon) \cap \Gamma = \{e\}$.

The quotient space $X = \Gamma \backslash G$ is the set of left cosets $x = \Gamma g$.

Quotient metric: $d_{\Gamma \backslash G}(x, x') = \inf_{\gamma \in \Gamma} d_G(g, \gamma g')$

Let $p: G \rightarrow X$ be the projection map. $\varepsilon = \varepsilon(g)$

lem: $\forall g \in G$, the map p induces a homeo. from

$B(g, \varepsilon) \rightarrow B(p(g), \varepsilon)$. And $p^{-1}B(p(g), \varepsilon)$ is a disjoint union of balls $B(\gamma g, \varepsilon)$.

pf. Let $\varepsilon(g) = \frac{1}{3} \inf_{\gamma \neq \text{id} \in \Gamma} d(g, \gamma g)$

$= \frac{1}{3} \inf_{\substack{\gamma \in \Gamma, \\ \gamma \neq \text{id}}} d(e, g^{-1}\gamma g)$ $g^{-1}\Gamma g$ discrete subgroup

$d(\gamma' g, \gamma g) = d(g, (\gamma'^{-1}\gamma)g) \geq 3\varepsilon(g)$

$\Rightarrow B(\gamma' g, \varepsilon), B(\gamma g, \varepsilon)$ disjoint.

lem. $\exists$ a Borel measure μ on X s.t. right G -inv.

i.e. $\forall g \in G, e(g)_* \mu = \mu$ second countable

pf: First notice ~~$SL_N(\mathbb{Q})$ dense in~~ $SL_N(\mathbb{R})$. $\forall g \in SL_N(\mathbb{Q})$ $\mathbb{R}$

$\exists \varepsilon(g)$ s.t. $B(g, \varepsilon(g)) \rightarrow B(p(g), \varepsilon(g))$ homeo.

~~We get a countable covering of~~ $X = \Gamma \backslash G \subset \bigcup_{g \in SL_N(\mathbb{Q})} B(p(g), \varepsilon(g))$

$\exists B_i$ in the countable basis of G .

$g \in B_i \subset B(g, \varepsilon(g))$.

Then get a family $G \subset \bigcup B_i$, each B_i injective

Def: A discrete subsp Γ of G is a lattice if (4)
 $\mu(\Gamma \backslash G) < \infty$. cocompact (uniform) if $\Gamma \backslash G$ compact.

Examples. (1) - $SL_N(\mathbb{Z})$ is a non-cocompact lattice of $SL_N(\mathbb{R})$
 (2) - $SL_N(\mathbb{Z})$ also contains cocompact lattices

Unitary representations

Def: Let G be a locally cpt gr and $\mathcal{H}$ a Hilbert space
 Let $U(\mathcal{H})$ be the group of unitary operators on $\mathcal{H}$.

That is $u \in U(\mathcal{H})$ linear operators on $\mathcal{H}$ s.t.
 $u^* u = u u^* = \text{id}$. (i.e. $(u(v), u(w)) = (v, w)$
 $\forall v, w \in \mathcal{H}$)

Def: A unitary rep of G on $\mathcal{H}$ is a morphism

$\pi: G \rightarrow U(\mathcal{H})$ s.t. the map $\forall v \in \mathcal{H}$

$G \mapsto \pi(g)v$ is continuous from $G \rightarrow \mathcal{H}$.

Example, $G \curvearrowright \mathcal{H} = L^2(\Gamma \backslash G, d\mu) \hookleftarrow G \quad \forall \varphi \in \mathcal{H}$

$$(\pi(g)\varphi)(x) = \varphi(xg)$$

is a unitary representation

Pf, $\pi(g)$ preserves the measure μ . s.t

$$\begin{aligned} \langle \pi(g)\varphi, \pi(g)\psi \rangle &= \int \varphi(xg) \overline{\psi(xg)} d\mu(x) \\ &= \int \varphi \overline{\psi} d\mu = (\varphi, \psi) \end{aligned}$$

Continuity. $\forall \varphi. \exists \varphi_n \in C_c^\infty(X)$ s.t. $\varphi_n \xrightarrow{L^2} \varphi$ (5)
 $(X \text{ not loc cpt Hausdorff, } \mu \text{ Borel})$
 $\| \pi(g_m) \varphi - \pi(g) \varphi \|_{L^2} \leq 2 \| \varphi_n - \varphi \|_{L^2} + \| \pi(g_m) \varphi_n - \pi(g) \varphi_n \|_{L^2}$

$$\| \pi(g_m) \varphi_n - \pi(g) \varphi_n \|_{L^2} \xrightarrow{m \rightarrow \infty} 0$$

dominated convergent
uniform continuity

$$d(xg, x) = \inf_{g_0 \in T} d(g_0 g, g_0) \leq d(g_0 g, g_0) = d(g, e)$$

So $\text{supp } \pi(g_m) \varphi_n$ is contained in $\text{supp } \pi(g) \varphi_n$

a neighborhood of $\text{supp}(\pi(g) \varphi_n)$ with size $d(g, g_m)$.

All functions $(\pi(g_m) \varphi_n)_m$ support on a same cpt set. $\square$

Example. $L_0^2(X, \mu) = \{ \varphi \in L^2(X, \mu) \mid \int \varphi d\mu = 0 \}$

G invariant subspace of $\mathcal{L}$.

and $L_0^2(X, \mu)^G = \{ 0 \}$, G -invariant function

pf. By definition. $\forall g \in G. \varphi(xg) = \varphi(x)$ for μ -a.e. x

Then for μ -a.e. x $\varphi(xg) = \varphi(x)$ for ν -a.e. g

Fix a such x . then $\varphi(xg) \neq \varphi(x)$ for a g in a set of ν measure 0.

Then such xg also has μ measure 0

Hence $\varphi(y) = \varphi(x)$ for μ -a.e. y

$$\begin{array}{c} \nu(B(g, \varepsilon)) \\ \downarrow \\ \mu(xB(g, \varepsilon)) \end{array}$$

$\square$

Def. Matrix coefficient for $v, v' \in \mathcal{H}$.

(6)

and $g \in G$. let

$$C_{v,v'}(g) = \langle \pi(g)v, v' \rangle$$

~~$\forall h, h' \in G$.~~

$$\langle \pi(h) \pi(h') v, v' \rangle = \langle \pi(h'gh) v, v' \rangle$$

If $\pi(h)v = v, \pi(h')v' = v'$, then

$$\begin{aligned} (\star) \quad C_{v,v'}(g) &= \langle \pi(g)v, v' \rangle = \langle \pi(g)h)v, \pi(h')^{-1}v' \rangle \\ &= C_{v,v'}(h'gh) \end{aligned}$$

Ergodicity of quotients of $SL_2(\mathbb{R})$

For $t > 0$, and $s \in \mathbb{R}$, let $a_t = \begin{pmatrix} t & \\ & t^{-1} \end{pmatrix}$, $u_s = \begin{pmatrix} 1 & s \\ 0 & 1 \end{pmatrix}$

$$u_s^{-1} = \begin{pmatrix} 1 & 0 \\ s & 1 \end{pmatrix}$$

Thm. Let Γ be a lattice in $SL_2(\mathbb{R})$. $t \neq 0$ ($s \neq 0$)

Then the action of a_t (u_s or u_s^{-1}) is ergodic for the measure μ .

Consequence of the proposition

Prop. For any vector v in $\mathcal{H} = L^2(\Gamma \backslash G, \mu)$.

v is a_t (u_s or u_s^{-1}) -inv. then v is G -invariant

lem. (Mautner) $\forall v \in \mathcal{H}$ let $G_v = \{g \mid \pi(g)v = v\}$

i). G_v is a closed subgroup

ii). $g \in G_v \iff C_{v,v}(g) = \|v\|^2$

$$\text{iii)} \quad g \in G \text{ s.t. } \exists g_n \in G, s_n, s_n' \in G_v$$

(7)

$$\text{with } \lim g_n = g, \quad \lim s_n g_n s_n' = e.$$

$$\text{Then } g \in G_v$$

Pf. i) by continuity

$$\text{ii)} \quad |C_{v,v}(g)| = |\langle \pi(g)v, v \rangle| \leq \|v\|^2.$$

$$\|\pi(g)v - v\|^2 = 2\|v\|^2 - 2\operatorname{Re}(\langle \pi(g)v, v \rangle)$$

$$\pi(g)v = v \Rightarrow \operatorname{Re}(\langle \pi(g)v, v \rangle) = \|v\|^2 \Rightarrow \langle \pi(g)v, v \rangle = \|v\|^2 \\ \Leftarrow \text{trivial}$$

$$\text{iii)} \quad \pi(g)v = \lim \pi(g_n)v$$

by *

$$= \lim \pi(s_n g_n s_n') v = \pi(e)v = v \\ C_{v,v}(g) = \lim C_{v,v}(g_n) = \lim C_{v,v}(s_n g_n s_n') \\ = C_{v,v}(e) = \|v\|^2$$

pf of Prop:

$$\text{By ii)} \Rightarrow g \in G_v$$

Ac-inv.

$$\text{let } g_n = u_s.$$

$$s_n = a_{t^n}.$$

$$s_n' = a_{t^{-n}}.$$

$$\begin{pmatrix} t & \\ & t^0 \end{pmatrix} \begin{pmatrix} 1 & s \\ & 1 \end{pmatrix} \begin{pmatrix} t & \\ & t^{-1} \end{pmatrix} = \begin{pmatrix} 1 & st^{-2} \\ & 1 \end{pmatrix}$$

Suppose $t > 0$.

$$s_n g_n s_n' = a_{t^n} u_s a_{t^{-n}} = u_s t^{-2n} \xrightarrow{n \rightarrow \infty} e.$$

By lem $u_s \in G_v$.

u_s similarly

$$\forall s \in \mathbb{R}$$

$$\underline{U_s - \text{inv}} \Rightarrow A_t - \text{inv} \text{ for } \forall t > 0$$

(8)

pf. It is sufficient to prove ^{case} $t = \frac{p}{q}$, rational

let
$$g_n = \begin{pmatrix} \frac{p}{q} & \\ \frac{-t+1}{snp} & \frac{q}{p} \end{pmatrix} \quad g = A_t$$

$$S'_n = U_s^{-np} = \begin{pmatrix} 1 & -snp \\ & 1 \end{pmatrix} \quad S_n = U_s^{np} = \begin{pmatrix} 1 & snp \\ & 1 \end{pmatrix}$$

Then
$$S_n g_n S'_n = \begin{pmatrix} 1 & 0 \\ \frac{-t+1}{snp} & 1 \end{pmatrix} \xrightarrow{n \rightarrow \infty} \mathbb{Q}$$

Finally. For $A_t - \text{inv}$ (U_s or $U_s^- - \text{inv}$), we get

A_t, U_s, U_s^- inv for $\forall t \in \mathbb{R} > 0, s \in \mathbb{R} \in G_v$

These three subgroups generate $SL_2(\mathbb{R})$.

Hence $G_v = G$.

考试内容到 mixing 为止, 后面为拓展内容, 不同类型的动力系统